Leonardo Borgioli

PhD Candidate in AI for Robotic-Assisted Surgery

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Profile

PhD candidate in video and multimodal foundation models, self-supervised learning, and robotics. Experience training billion-parameter models on leadership-class HPC systems for surgical and scientific applications. Expected graduation: May 2027.

Professional Experience

Research Intern – AI and High-Performance Computing — **Argonne National Laboratory**
Lemont, IL — *05/2026–present*

- Developed OmniRAS, 1B/2B V-JEPA 2.1 video foundation models pretrained on approximately 2,650 hours of surgical video and evaluated across six tasks: phase/step/triplet recognition, action segmentation, detection, and cross-domain transfer.
- Scaled V-JEPA 2/2.1 pretraining from ViT-L/ViT-H to billion-parameter architectures, including a 36.9M-clip run on Argonne Leadership Computing Facility systems.
- Conducted 250+ downstream experiments on representation transfer, partial-backbone fine-tuning, label-free model selection, and scaling behavior across surgical benchmarks.
- Developed a multimodal JEPA architecture combining equilibrium video with heterogeneous plasma diagnostic and control signals through cross-attention for tokamak plasma dynamics.

Graduate Research Assistant — **Surgical Innovation and Training Laboratory, UIC**
Chicago, IL — *05/2023–present*

- Developed RASRAG, a hierarchical retrieval-augmented generation framework and benchmark for robot-assisted surgery, evaluated with surgeons and physicians.
- Developed a sensory-glove interface for the da Vinci Research Kit, achieving console-comparable accuracy and usability after approximately one hour of surgeon training.
- Co-created CRCD, a multimodal dataset of 21 robotic procedures and >120,000 annotations spanning video, kinematics, pedal signals, and surgical labels.
- Developed autonomous-dissection systems for deformable surgical environments, achieving approximately 0.5-mm tracking precision.

Graduate Teaching Assistant — **University of Illinois Chicago**
Chicago, IL — *08/2022–05/2023*
ECE 350: Principles of Automatic Control; ECE 452: Robotics: Algorithms and Control.

Master's Thesis Research Intern — **Italian Institute of Technology**
Genoa, Italy — *01/2022–07/2022*
Developed mixed-reality interfaces and compliant-control methods for upper-limb robotic rehabilitation.

Consultant Intern – Neural Network Study — **DigiSky**
Turin, Italy — *06/2021–08/2021*
Studied neural-network methods for classification and segmentation of aerial imagery.

Education

PhD in Electrical and Computer Engineering — **University of Illinois Chicago**
Chicago, IL — *08/2022–present*
Foundation models and autonomy for robot-assisted surgery. Preliminary examination passed May 2026; expected graduation May 2027.

EMARO+ Double Master's Degree in Advanced Robotics — **École Centrale de Nantes and Università di Genova**
France and Italy — *2020–2022*
European double degree in robotics, AI, automation, and computer science. GPA: 3.9/4.0.

Bachelor's Degree in Electronics, Electrical Engineering, and Automation — **Sorbonne University**
Paris, France — *2017–2020*

Technical Skills

- **AI and Foundation Models:** PyTorch, V-JEPA 2/2.1, video transformers, self-supervised learning, multimodal fusion, RAG, PEFT.
- **Vision and Robotics:** DINOv2, SAM 2, YOLO, Detectron2, temporal action segmentation, dVRK, ROS.
- **Programming and HPC:** Python, C/C++, CUDA, distributed training, Aurora, Polaris, Sophia, PBS, Linux, Git.
- **Languages:** English, French, Italian.

Publications

Highlighted Publications

- OmniRAS — First author: L. Borgioli, N. Getty, W. Xiu, J. Cassiani, A. Ducas, C. A. Orda, H. Waris, F. Xia, R. Stevens, P. C. Giulianotti, and M. Žefran. “OmniRAS: Standardizing Foundation Model Training and Evaluation in Robot-Assisted Surgery.” Submitted to Medical Image Analysis (MedIA). arXiv:2608.31048.
- SC26 Workshop (2026) — First author, accepted: L. Borgioli, N. J. Getty, M. Žefran, and S. Madireddy. “On Multimodal Joint Embedding Predictive Architecture models for Tokamak Plasma Dynamics.”
- IROS 2026 — First author: L. Borgioli, K.-H. Oh, A. Duncas, N. Getty, M. Žefran, and P. C. Giulianotti. “When Self-Supervision Transfers: Quantifying the Laparoscopic–Robotic Domain Shift.”

Peer-Reviewed Conference Papers

- SC26 Workshop (2026) – Accepted: M. Siddique, N. Getty, and L. Borgioli. “Out-of-View Depth and Motion Prediction in Robotic Endoscopy.”
- IROS 2025: K.-H. Oh, L. Borgioli, M. Žefran, V. Valle, and P. C. Giulianotti. “Autonomous Dissection in Robotic Cholecystectomy.”
- ICRA 2025: L. Borgioli, K.-H. Oh, V. Valle, A. Ducas, M. Halloum, D. F. M. Medina, A. Sharifi, et al. “Sensory Glove-Based Surgical Robot User Interface.”
- ISMR 2024: K.-H. Oh*, L. Borgioli*, A. Mangano, V. Valle, M. Di Pangrazio, F. Toti, G. Pozza, et al. “Comprehensive Robotic Cholecystectomy Dataset (CRCD): Integrating Kinematics, Pedal Signals, and Endoscopic Videos.” (*Equal contribution)
- BioRob 2024: K.-H. Oh, L. Borgioli, M. Žefran, L. Chen, and P. C. Giulianotti. “A Framework for Automated Dissection Along Tissue Boundary.”

Peer-Reviewed Journal Articles

- Science Robotics (2026) – Accepted: Open-H-Embodiment Consortium, including L. Borgioli. “Open-H-Embodiment: A Large-Scale Dataset for Enabling Foundation Models in Medical Robotics.” arXiv:2604.21017.
- Surgery (2026): T. Marchese, V. Valle, B. Ielpo, M. G. Spampinato, A. Comandatore, L. Borgioli, P. C. Giulianotti, L. Morelli, and the PANFRAIL Collaborative Group. “Robotic Pancreatoduodenectomy in Elderly Individuals: An International Multicenter Propensity Score–Matched Study by the PANFRAIL Collaborative Group.” Article 110153.
- BMC Surgery (2026): W. Xiu, J. Liu, J. Chen, J. Zhang, C. Yang, K. Wang, R. Zhu, L. Yu, X. Zhou, F. Wang, L. Borgioli, A. Ducas, S. Jia, and Q. Dong. “Patient-Specific Virtual Surgical Simulation: The Preliminary Exploration of Surgical Evaluation for Difficult Pediatric Liver Tumors.” Vol. 26, Article 44. Published online November 28, 2025.
- Journal of Medical Robotics Research (2025): K.-H. Oh*, L. Borgioli*, A. Mangano, V. Valle, M. Di Pangrazio, F. Toti, G. Pozza, et al. “Expanded Comprehensive Robotic Cholecystectomy Dataset (CRCD).” Vol. 10, Nos. 3–4, Article 2550006. (*Equal contribution)
- Annals of Hepato-Biliary-Pancreatic Surgery (2025): A. Ducas, A. Mangano, L. Borgioli, J. Cassiani, P. Lopez, and P. C. Giulianotti. “The Role of Artificial Intelligence in Pancreatic Surgery: Current and Future Perspectives.” Vol. 29, No. 1, pp. 1–4.

Manuscripts Under Review

- ICRA 2027 – Submitted: L. Borgioli, et al. “Toward Consistent Multi-Annotator Labeling of Concurrent Surgical Action Triplets.”
- ICRA 2027 – Submitted: L. Borgioli, et al. “Label-free Metrics for Selecting Surgical Video Representations.”
- L. Borgioli, K.-H. Oh, A. Ducas, W. Xiu, et al. “RASRAG: A Domain-Specific RAG Framework and Benchmark for Robotic-Assisted Surgery.” Under review at the IEEE Journal of Biomedical and Health Informatics.

Invited Talks and Presentations

- Invited Faculty Presentation – CRSA Worldwide Congress (2025): Clinical Robotic Surgery Association, Los Angeles, CA, USA. “Large Language Models in Robotic-Assisted Surgery: Methodologies and Unresolved Technical Challenges.”
- Invited Talk – Hospital del Mar: Barcelona, Spain. “Large Language Models for Data Science.”.

Posters

- CRSA 2023: L. Borgioli, et al. “Feasibility Study of Control of the dVRK Using HTC Vive Pro Controller.”
- CRSA 2023: L. Borgioli, et al. “Towards Autonomous Clutch in Robotic Surgery.”

Awards

- Best Poster Award, CRSA 2023: “Feasibility Study of Control of the dVRK Using HTC Vive Pro Controller.”